Wenbin Teng

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Education

University of Southern California — Los Angeles, CA
Ph.D. in Computer Science, Advisor: Yajie Zhao — *Aug. 2022 – Exp. Dec. 2027*

Boston University — Boston, MA
B.A. in Statistics, B.A. in Economics, Magna Cum Laude, GPA: 3.93 — *Sep. 2015 – Jan. 2019*

Xi'an Jiaotong University — Xi'an, Shaanxi, China
Concentration in Finance, GPA: 83.8/100 — *Sep. 2013 – July 2015*

Research Areas

Video Generation, World Modeling, Diffusion Model, Auto-regressive Model, Novel View Synthesis

Preprints and Publications

STRIKE: Learning Visual State Transitions for Physical World Modeling — In Submission
***Wenbin Teng**, Tianshuo Xu, Depu Meng, Yuelei Li, Quentin Herau, Yihan Hu, Yajie Zhao, Wei Zhan* — *2026*

XDG: Accelerated Visual Disambiguation — WACV
*Gonglin Chen, Ben Southall, Hanyuan Xiao, **Wenbin Teng**, Haolin Xiong, Tianwen Fu, Junyi Ouyang, Kshitij Singh Minhas, Supun Samarasekera, Rakesh Kumar, and Yajie Zhao* — *2027*

DCARL: A Divide-and-Conquer Framework for Autoregressive Long-Trajectory Video Generation — ECCV
*Junyi Ouyang, **Wenbin Teng**, Gonglin Chen, Haiwei Chen, Yajie Zhao* — *2026*

ARSS: Taming Decoder-only Autoregressive Visual Generation for View Synthesis From Single View — Preprint
***Wenbin Teng**, Gonglin Chen, Haiwei Chen, Yajie Zhao* — *2025*

FVGen: Accelerating Novel-View Synthesis with Adversarial Video Diffusion Distillation — ICCV
***Wenbin Teng**, Gonglin Chen, Haiwei Chen, Yajie Zhao* — *2025*

RDD: Robust Feature Detector and Descriptor using Deformable Transformer — CVPR
*Gonglin Chen, Tianwen Fu, Haiwei Chen, **Wenbin Teng**, Hanyuan Xiao, Yajie Zhao* — *2025*

Skyeyes: Ground Roaming with Aerial View Images — WACV
Zhiyuan Gao, **Wenbin Teng***, Gonglin Chen, Jinsen Wu, Ningli Xu, Rongjun Qin, Andrew Feng, Yajie Zhao* — *2025*
* *denotes equal contributions*

Geometry-aware Feature Matching for Large-Scale Structure from Motion — 3DV Oral
*Gonglin Chen, Jinsen Wu, **Wenbin Teng**, Zhiyuan Gao, Andrew Feng, Rongjun Qin, Yajie Zhao* — *2025*

Mind-to-Face: Neural-Driven Photorealistic Avatar Synthesis via EEG Decoding — ECCV Spotlight
*Haolin Xiong, Tianwen Fu, Pratusha Bhuvana Prasad, Yunxuan Cai, Haiwei Chen, **Wenbin Teng**, Hanyuan Xiao, Yajie Zhao* — *2025*

Light Sampling Field and BRDF Representation for Physically-based Neural Rendering — ICLR
*Jing Yang, Hanyuan Xiao, **Wenbin Teng**, Yunxuan Cai, Yajie Zhao* — *2023*

Unimodal Face Classification with Multimodal Training — IEEE FG
***Wenbin Teng**, Chongyang Bai* — *2021*

MVDG : Efficient Multi-view 3D Disambiguation on Unconstrained Real-World Images — Preprint
*Hanyuan Xiao, Gonglin Chen, Haolin Xiong, **Wenbin Teng**, Haiwei Chen, Yajie Zhao* — *2026*

Expression-Diverse References for Identity-Preserving Video Generation — Preprint
*Tianwen Fu, **Wenbin Teng**, Gonglin Chen, Junyi Ouyang, Haolin Xiong, Yajie Zhao* — *2026*

Experience

Research Intern — May. 2026 – Present
Applied Intuition. Advisor: Dr. Wei Zhan — *Sunnyvale, CA*

Research Assistant — Aug. 2022 – Present
USC Institute for Creative Technologies. Advisor: Dr. Yajie Zhao — *Los Angeles, CA*

- Lead research projects on scene construction and sparse novel view synthesis with diffusion model.
- Participated in research projects related to feature matching, Structure from Motion Reconstruction, NeRF and 3D Gaussian Splatting.

Research Intern — May 2024 – Aug 2024
Center for Vision Technologies, SRI International. Advisor: Dr. Supun Samarasekera — *Princeton, NJ*

- Research on sparse novel view synthesis with multi-view and video diffusion models.
- Research on sparse 3D reconstruction with customized 3D Gaussian Splatting.

Computer Vision Engineer Intern — Oct 2020 – Mar 2021
Department of Computer Vision Technology, Baidu Inc. Advisor: Dr. Haocheng Feng — *Beijing, China*

- Proposed an image-to-image translation framework for unsupervised domain adaptation on cross-database face anti-spoofing.

Research Intern — May 2020 – Aug 2021
CCVL Lab, John Hopkins University. Advisor: Prof. Alan Yuille — *Baltimore, MD*

- Researched on model-based face autoencoder for 3D face rendering.
- Performed iterative adversarial attack on the autoencoder through perturbation optimization.

Research Intern — Jan 2020 – Aug 2021
Computer Science Department, Dartmouth College. Advisor: Dr. Chongyang Bai — *Hanover, NH*

- Proposed a multi-modal training and uni-modal testing (MTUT) framework on face classification task.

Research Assistant — Jan 2018 – Aug 2018
Department of Statistics, Boston University. Advisor: Allen G. Harbaugh — *Boston, MA*

- Conducted research on patterns of model selection protocols such as Mallow's *Cp* and information criteria with simulated data.

Project

3D MRI Image Translation with Diffusion Model — USC
Course Project – Probabilistic and Generative Models — *Spring 2025*

GaussianNeRF: High-Fidelity Surface Reconstruction based on Gaussian Constraint — USC
Course Project – Advanced Computer Vision — *Fall 2023*

Score-based Generative Modeling with Stochastic Differential Equations — USC
Course Project – Mathematics of High-Dimensional Data — *Fall 2023*

Parallel Globally Consistent Normal Orientation of Raw Unorganized Point Clouds — USC
Course Project – Geometric Shape Modeling — *Spring 2024*

Modeling Subsurface Scattering for Human Face Skin — USC
Course Project – 3D Graphics and Rendering — *Fall 2022*

Academic Services

WACV 2025
WACV Workshop 2025
CVPR 2025
CVPR Workshop on Synthetic Data for Computer Vision 2025
CVPR Workshop on Synthetic Data for Computer Vision 2026
NeuralIPS 2026
BMVC 2026
ICLR 2027

Teaching

Introduction to Programming in C++ — USC
Teaching Assistant — *Fall 2022*

Database Systems — USC
Teaching Assistant — *Spring 2023, Summer 2025, Spring 2026*

Introduction to Artificial Intelligence — USC
Teaching Assistant — *Fall 2024*

Machine Learning for Data Science — USC
Teaching Assistant — *Fall 2025*

Skills & Interests

Programming: Python, C++, Java, SQL, R, LaTeX
Frameworks: PyTorch, Tensorflow, PaddlePaddle
Cloud & DevOps: AWS, Git, Docker, Linux
Languages: Mandarin (Native), English (Fluent)
Interests: Badminton, Fitness, Cooking